Quasi-Poisson and Negative Binomial models

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Overdispersion

Overdispersion occurs when the response variable Y_i has greater variability than the model accounts for.

Poisson: $Var(Y_i|\mathbf{x}_i) = \mathbb{E}[Y_i|\mathbf{x}_i] = \mu_i$

If we fit a Poisson regression model, overdispersion means that $Var(Y_i|\mathbf{x}_i) > \mu_i$

Behavior of estimated coefficients in a mis-specified model

Suppose our systematic component is right: $\log(\mu_i) = x_i^T\beta$

But random component is wrong: $Y_i|x_i$ is not Poisson

If the model were correct: $\sqrt{n}(\hat{\beta}-\beta) \xrightarrow{d} N(0, \mathcal{I}_1^{-1}(\beta))$

$\mathcal{I}_1(\beta) = \mathbb{E}[\mu_i x_i x_i^T]$ for Poisson

If distribution is wrong:

$\hat{\beta}$ solves $\sum_{i=1}^{n}(Y_i-\mu_i)x_i = 0$ => M estimator!

Under mild conditions (do not require correct model!)

$$\sqrt{n}(\hat{\beta}-\beta) \xrightarrow{d} N(0, S)$$

$S = A^{-1}BA^{-1}$

$A = -\mathbb{E}\left[\frac{\partial}{\partial\beta}((Y_i-\mu_i)x_i)\right] = -\mathbb{E}[\mu_i x_i x_i^T]$

$B = \mathrm{Var}((Y_i-\mu_i)x_i)$

$= \mathbb{E}[(Y_i-\mu_i)^2 x_i x_i^T]$

$= \mathbb{E}[\mathrm{Var}(Y_i|x_i)\, x_i x_i^T]$

$$S = (\mathbb{E}[\mu_i x_i x_i])^{-1}\,\mathbb{E}[\mathrm{Var}(Y_i|x_i)\, x_i x_i^T]\,(\mathbb{E}[\mu_i x_i x_i^T])^{-1}$$

Behavior of estimated coefficients in a mis-specified model

$$S = \left(\mathbb{E}[\mu_i x_i x_i^T]\right)^{-1} \mathbb{E}[\mathrm{Var}(Y_i|x_i)\, x_i x_i^T] \left(\mathbb{E}[\mu_i x_i x_i^T]\right)^{-1}$$

Now: Suppose $\mathrm{Var}(Y_i|x_i) = \phi\mu_i$

Then: $S = \phi \left(\mathbb{E}[\mu_i x_i x_i^T]\right)^{-1}$

$= \phi$ (Poisson Fisher info)$^{-1}$

So: if we're worried about overdispersion:

- Option 1: use sandwich variance! (approximately correct)
- Option 2: <u>if</u> we believe $\mathrm{Var}(Y_i|x_i) = \phi\mu_i$ sandwich variance simplifies

⋮

quasi-Poisson models

$$\log(\mu_i) = \mathbf{x}_i^T \beta$$
$$Var(Y_i|\mathbf{x}_i) = \phi\mu_i$$

- Estimated coefficients ($\widehat{\beta}$) are the same
- Different estimated variance-covariance matrix

$$\widehat{Var}(\hat{\beta})_{QP} = \hat{\phi}\ \widehat{Var}(\hat{\beta})_{Poisson}$$

Just M-estimation using the usual Poisson score, + assumption on $Var(Y_i|X_i)$!

Example: articles data

```
glm(art ~ phd + ment + kid5, family = poisson, data = artic
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.276493   0.087294   3.167  0.00154 **
phd          0.008071   0.026358   0.306  0.75944
ment         0.026537   0.002001  13.264  < 2e-16 ***
kid5        -0.099696   0.035248  -2.828  0.00468 **
(Dispersion parameter for poisson family
   taken to be 1)
```

```
glm(art ~ phd + ment + kid5, family = quasipoisson, data =
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *
(Dispersion parameter for quasipoisson family
   taken to be 1.88461)
```

Activity: comparing standard errors

Work on the class activity (course website).

Activity

Poisson output:

	Estimate	Std. Error	z value	Pr(>|z
(Intercept)	0.276493292	0.087294456	3.1673637	1.538278e-0
phd	0.008071036	0.026357684	0.3062119	7.594433e-0
ment	0.026537317	0.002000733	13.2637987	3.753884e-4
kid5	-0.099696463	0.035248105	-2.8284205	4.677832e-0

95% confidence interval for e^{β_3}:

for β_3: $-0.997 \pm 1.96\,(0.03525)$

for e^{β_3}: $\exp(-0.169), \exp(-0.031)$

$= (0.845, 0.961)$

Activity: comparing standard errors

Options for estimating $SE(\widehat{\beta}_j)$ when we fit a Poisson model:

- Estimate from the original Poisson model (may be wrong!)
- Bootstrap resampling (should be approximately correct even if model is misspecified)
- Sandwich variance (should be approximately correct even if model is misspecified)
- quasi-Poisson (inflate standard errors by $\sqrt{\widehat{\phi}}$) ⇐ approximately correct if $\text{Var}(Y_i|y_i) = \phi\mu_i$ (somewhat strong assumption)

Activity: comparing standard errors

- Poisson estimate: $\widehat{SE}(\hat{\beta}_3)_{\text{poisson}} = 0.03525$
- Bootstrap estimate: $\widehat{SE}(\hat{\beta}_3)_{\text{boot}} = 0.0490$
- Sandwich estimate: $\widehat{SE}(\hat{\beta}_3)_{\text{sandwich}} = 0.0483$
- quasi-Poisson estimate: $\widehat{SE}(\hat{\beta}_3)_{\text{quasi-poisson}} = 0.0484$

How do the estimates compare? What will be the effect (on confidence intervals and hypothesis tests) of incorrectly assuming a Poisson distribution when the data are overdispersed?

If overdispersion:

$\hat{SE}_{\text{poisson}}$ is too small

↳ CIs are too narrow ("95%" CI does not have 95% coverage)

↳ test stats. are too far from 0 (inflated type I error rate)